

Project Title

A Data Warehouse for Analyzing Energy Security in Europe Under Geopolitical Risk

Group Members

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Chosen Type of Project

Data Warehousing

Datasets

The project will use the following datasets:

1. Eurostat - Energy imports dependency.
Energy import dependency rate by country and energy product.
2. Eurostat / data.europa.eu - Emergency oil stocks in days equivalent
Monthly emergency oil stock data for European countries.

https://ec.europa.eu/eurostat/databrowser/view/nrg_ind_id/default/table?lang=en

https://ec.europa.eu/eurostat/databrowser/view/nrg_stk_oem/default/table?lang=en

3. Eurostat - Natural gas prices
Natural gas prices for household and non-household consumers.

https://ec.europa.eu/eurostat/databrowser/view/nrg_pc_202/default/table?lang=en

4. Eurostat - Electricity prices
Electricity prices for household and non-household consumers.

https://ec.europa.eu/eurostat/databrowser/view/nrg_pc_204/default/table?lang=en

5. Caldara-Iacoviello Geopolitical Risk Index
Monthly and daily geopolitical risk indicators.

<https://www.matteoiacoviello.com/gpr.htm>

Project Presentation

The project aims to design and implement a Data Warehouse in PostgreSQL to analyze European energy security during geopolitical crises. Heterogeneous datasets on import dependency, emergency oil stocks, energy prices, and geopolitical risk will be extracted, cleaned, and integrated through Python/SQL ETL scripts. Particular attention will be paid to temporal and spatial granularity alignment, aggregating monthly, semi-annual, and annual data when needed. A Dimensional Fact Model will be designed and translated into a star schema. OLAP queries will be used to compare EU countries and analyze how energy prices, import dependency, and emergency stock levels evolve during periods of high geopolitical risk.